REVIEW

# When the weekly signal fades


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*Stopping semaglutide is less like flipping a switch than watching an external biological assist recede. The evidence points to returning appetite pressure, frequent weight regain, and erosion of some metabolic gains—but not to one inevitable timetable.*

Twenty weeks into a large obesity trial, every participant had already received semaglutide. Then chance divided them: some kept taking it, while others were switched to placebo. Over the next 48 weeks, the continuing group lost another 7.9% of body weight, while the switch group gained 6.9%.[1] The two paths opened from the same pharmacological starting line.

That split is the cleanest answer to what happens after the final injection. Semaglutide does not usually leave behind a permanent reset. As exposure ebbs, the appetite and eating-control effects demonstrated during treatment can ebb too; weight often rises; and several improvements tied to weight and treatment move back toward baseline. Yet "often" is not "always," and the average curve is not a forecast for one body.

This direct withdrawal dataset followed participants from a semaglutide 2.4-mg obesity trial that excluded type 1 or type 2 diabetes.[2] That boundary matters. So does the difference between stopping under a trial protocol and stopping because a pharmacy cannot fill a prescription.

## 01 FIRST, THE DRUG LEAVES SLOWLY

Semaglutide was engineered to bind reversibly to albumin, which prolongs its action in the circulation.[3] Its long half-life supports once-weekly dosing.[4] A missed Monday dose therefore does not mean that the molecule has vanished by Tuesday.

The first change is a declining drug signal, not a special "withdrawal syndrome" established by clinical trials. Existing pharmacokinetic evidence cannot tell a person the exact day when hunger, blood sugar, or gastrointestinal sensations will change.

During treatment, small randomized studies repeatedly find lower ad-libitum food intake and better control of eating. One injectable-semaglutide study reported reduced appetite, food cravings, and energy intake at 20 weeks.[5] An earlier injectable study likewise found lower energy intake after 12 weeks.[6] In adults with obesity, an oral-semaglutide experiment found reduced energy intake and appetite.[7] In adults with type 2 diabetes, another oral-semaglutide experiment found greater satiety and fullness.[8] A systematic review reached the same broad conclusion across GLP-1 analogues.[9]

Those are on-treatment findings. They make the return of appetite pressure biologically plausible, but the major withdrawal trials did not continuously measure hunger from the final dose onward. They also complicate a popular explanation. One 20-week injectable-semaglutide study found no detectable delay in gastric emptying by paracetamol absorption at that time point.[5] A 20-week oral-semaglutide study reported the same null finding at that time point.[7]

## 02 THE SCALE USUALLY TURNS BEFORE IT SETTLES

Semaglutide's active-treatment record is unusually consistent. In STEP 1, participants assigned to 2.4 mg lost 14.9% of baseline weight by week 68 on average.[10] In STEP 5, the mean loss was 15.2% at week 104.[11] A long-term meta-analysis estimated a 12.3-kg advantage over placebo.[12] Another review found mean losses of 14.9% to 17.4% across four STEP trials.[13] A broader medication review estimated that semaglutide produced 11.4% more weight loss than placebo.[14]

Continuation preserves that pressure on weight; withdrawal removes it. The randomized STEP 4 divergence is not a before-and-after anecdote, because both arms had the same run-in and both retained lifestyle intervention.[1]

The randomized and extension trajectories are set side by side in Figure 1.

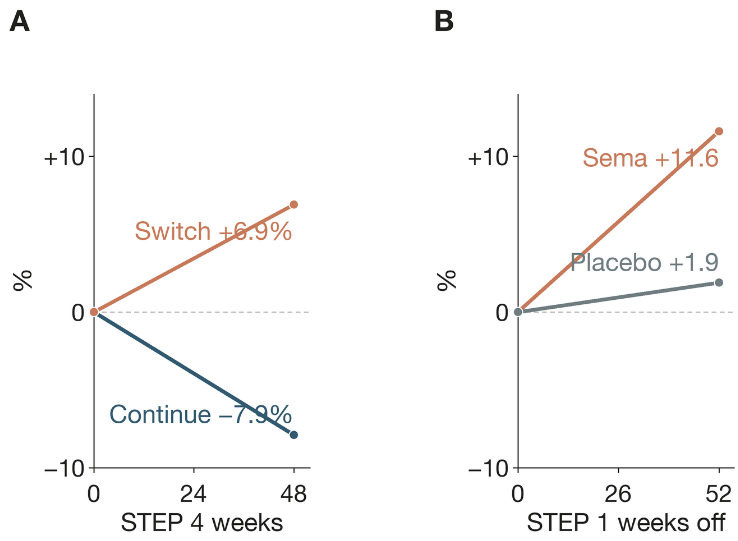


**Figure 1. Weight turns upward when treatment ends.** STEP 4 reports change after week-20 randomization.[1] "Sema" means semaglutide. The designs and denominators differ, so the panels should not be numerically pooled.

STEP 1 supplies the longer view. During its 52-week extension without study treatment, former semaglutide recipients regained 11.6 percentage points, leaving them 5.6% below their original baseline on average.[2] At that one-year mark, 48.2% still retained at least a 5% loss.[2] "Regain" therefore does not mean that every participant returned to their starting weight.

Across GLP-1-receptor-agonist trials, however, the direction is remarkably stable. One meta-analysis concluded that regain was proportional to the original loss.[15] A second found rebound across 11 randomized obesity trials but extreme between-study heterogeneity.[16] A third described rapid regain across GLP-1 and dual GIP/GLP-1 agents after cessation.[17]

## 03 A FAST AVERAGE IS NOT YOUR STOPWATCH

A 2026 synthesis estimated that newer incretin medicines were followed by regain of 0.8 kg per month during observed follow-up.[18] Its estimate is useful as a group-level pace, not a deadline. For the newest medicines, observed post-cessation data extended only to 12 months; later points in that analysis were projected.[18]

The reported pooled magnitudes and their uncertainty appear in Figure 2.

The variation is not statistical clutter to be wished away. It reflects different drugs, doses, treatment lengths, populations, and follow-up programs. STEP 1 subgroup patterns were inconsistent across starting-body-size categories.[2] A small polycystic-ovary-syndrome cohort also reported highly heterogeneous individual responses after semaglutide.[19] No validated calculator can yet turn those averages into your month-by-month curve.

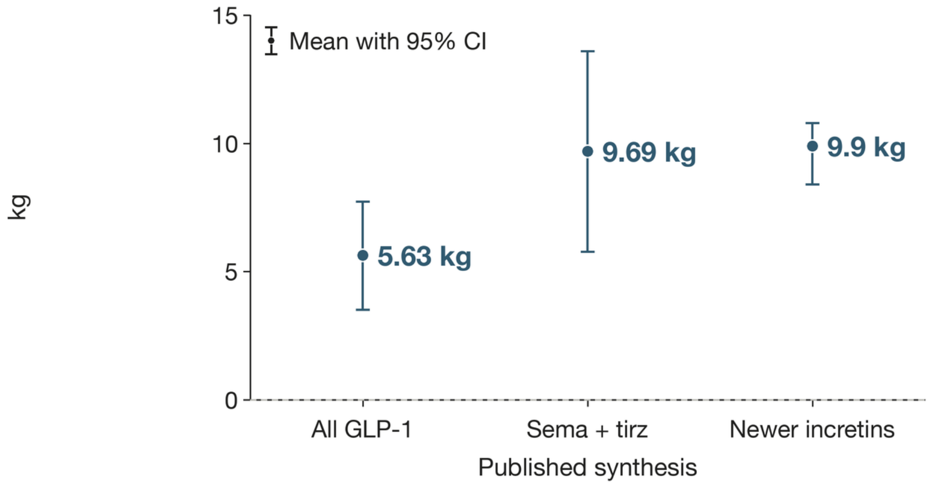


**Figure 2. Across 11 randomized trials in participants with overweight or obesity, GLP-1RA discontinuation was followed by 5.63 kg mean regain.** That plotted value is the meta-analysis's overweight-or-obesity stratum.[16] The "sema + tirz" estimate pools semaglutide and tirzepatide.[15] The newer-incretin estimate is a one-year model output.[18] Whiskers show reported 95% confidence intervals; these differently scoped points are not interchangeable personal forecasts.

| EVIDENCE WINDOW | WHAT CHANGED AFTER STOPPING | KEY LIMITATION OR BOUNDARY |
|---|---|---|
| STEP 4 randomized withdrawal | Switch-to-placebo participants gained 6.9% over 48 weeks | Only participants who reached the 2.4-mg weekly maintenance dose were randomized |
| STEP 1 extension | Former semaglutide participants regained 11.6 percentage points in one year[2] | Semaglutide and the lifestyle intervention both ended |
| Newer-incretin synthesis | Mean regain rate was 0.8 kg per month during observed follow-up | Observed follow-up ended at 12 months; later values were projections |
| PCOS cohort continuing metformin | Participants regained about one-third of prior loss over two years[19] | What would happen without metformin or in other populations |

## 04 MORE THAN WEIGHT MOVES

The body changes tracked after discontinuation are not confined to the scale. In STEP 4, waist circumference, systolic blood pressure, fasting glucose, fasting insulin, and lipids improved during the semaglutide run-in, stayed improved with continued treatment, and deteriorated after the switch to placebo.[20] In the STEP 1 extension, most cardiometabolic variables moved back toward baseline after withdrawal, though some net benefit remained at one year.[2] A modeled ten-year type 2 diabetes risk score likewise fell during treatment and rose after withdrawal.[21]

The direction is clearer than the clinical meaning for any individual. A meta-analysis estimated that glycated hemoglobin, or HbA1c—a months-long measure of average blood sugar—rose by 0.25 percentage points in obesity cohorts after GLP-1-agonist cessation.[16] The same analysis estimated a 0.65-point rise in type 2 diabetes cohorts.[16] Those pooled values were highly heterogeneous.

Continuous semaglutide reduced major cardiovascular events in 17,604 high-risk adults without diabetes in SELECT.[22] SELECT did not test whether that event protection persists after stopping. Marker rebound cannot fill that evidentiary gap.

## 05 OUTSIDE TRIALS, STOPPING HAS MANY MEANINGS

In everyday care, "stopped" may mean a planned transition, an intolerable adverse effect, an insurance denial, a shortage, or a long refill gap. Routine-care cohorts therefore report several different endpoints rather than one universal stopping rate.

Those endpoints are set side by side in Figure 3.

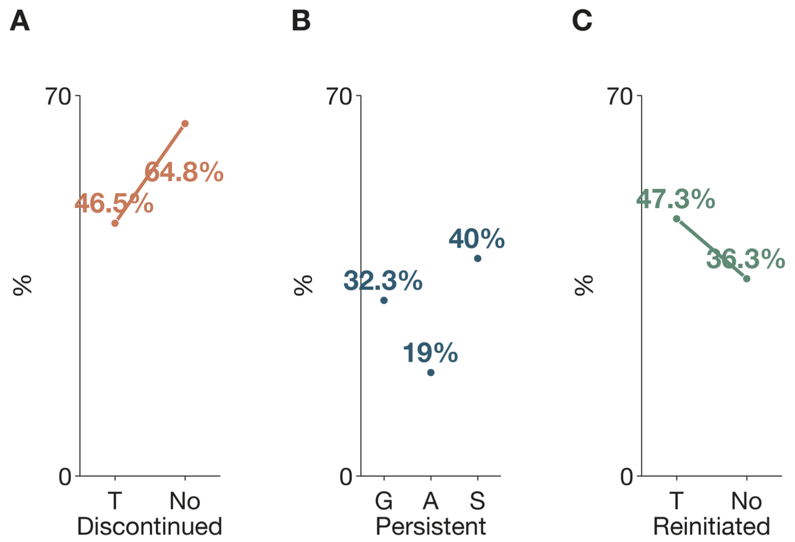


**Figure 3. In a 125,474-person cohort, one-year discontinuation was 46.5% with and 64.8% without type 2 diabetes; reinitiation was 47.3% and 36.3%, respectively.** The study stratified results by type 2 diabetes status.[23] A GLP-1 study reported 32.3% one-year persistence.[24] An anti-obesity-medication cohort reported 19% overall and 40% for semaglutide.[25] Panel B abbreviates these as "GLP," "All," and "sema." Definitions and denominators differ.

Why do people stop? In a 288-person chart review, cost or insurance problems accounted for 47.6% of documented first-year discontinuations; side effects accounted for 14.6%, and shortages for 11.8%.[26] In another clinical cohort, notes most often named adverse effects and cost.[23] These records capture access as much as biology.

Routine-care weight outcomes track exposure, but not with experimental control. One cohort reported one-year weight reductions of 3.6% with early discontinuation, 6.8% with late discontinuation, and 11.9% without discontinuation.[27] Another found a 5.5% loss with persistent coverage, compared with 1.8% among patients with fewer than 90 covered days.[28] Selection, dose, indication, and access can contribute to those gaps.

## 06 CAN ANYTHING MAKE THE LANDING GENTLER?

Here the evidence changes character. It becomes smaller, less semaglutide-specific, and more dependent on selected populations.

The strongest randomized clue comes from liraglutide, a different glucagon-like peptide-1 receptor agonist, or GLP-1 drug. People who combined supervised exercise with liraglutide during treatment maintained weight and fat loss better one year after both interventions ended than people who had used liraglutide alone.[29] Some regain still occurred.

In 25 women with polycystic ovary syndrome who continued metformin and lifestyle support, about one-third of semaglutide-induced weight loss returned over two years.[19] In a propensity-matched telemedicine cohort for type 2 diabetes, more than 70% of deprescribed participants maintained at least 5% weight loss at 12 months while continuing a carbohydrate-restricted nutrition program.[30] Neither cohort proves that its approach will generalize.

What about tapering, a particular diet, or a handoff to exercise? A review says the efficacy of proposed lifestyle measures still awaits well-designed randomized trials.[31] There is no broadly validated semaglutide off-ramp yet.

## 07 THE PRACTICAL QUESTION IS NOT "WILL I FAIL?"

The evidence does not support framing regain as a moral verdict. Semaglutide alters appetite and eating while exposure continues; removing that pharmacological help changes the conditions under which weight was lost. Continued treatment can maintain large average losses, but gastrointestinal events cause some trial participants to discontinue.[10] Real-world access creates another layer of interruption.

A safer question is: which benefits is the person trying to preserve, and what monitoring or replacement support fits that reason? Someone taking semaglutide for type 2 diabetes has a different glucose risk from someone using it only for obesity. A person stopping for nausea has a different problem from someone confronting a denied refill. The pooled evidence cannot decide an individual medication plan.

The honest forecast has two levels. At the population level, appetite control is likely to loosen, weight regain is common, and some metabolic markers tend to deteriorate. At the personal level, magnitude and timing remain uncertain—and the smaller studies that beat the average did so with continuing structure, not with a biological reset.



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